Timeouts retries and cancellation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete timeout and retry-policy record.

Keep caps, gaps, attempt counts, failure categories, cancellation boundaries, events and remote-outcome uncertainty together.

Keep one habit: Recall → W1–W3/W5 → independent timing case → project waiting policy → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Approval gates and idempotency. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from State machines for bounded agents. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Evaluation pipelines. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- **Timeout**s bound waiting; retries consume a finite budget. **Cancellation** should stop future work while accounting for operations already committed.

- Retry only declared transient failures. Use at most three attempts with waiting gaps 1 and 2 seconds; the caller waits at most 4 seconds for each attempt under the enforced local timeout model. Worst-case elapsed bound excluding overhead is 3*4+1+2=15 seconds.

Approval gates and idempotency

> **My remembered idea, example and correction**

State machines for bounded agents

> **My remembered idea, example and correction**

Evaluation pipelines

> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main virtual policy: at most 3 total attempts, each capped at 4 seconds of caller waiting, with gaps 1 and 2 seconds before attempts 2 and 3. Under enforced local caps and excluding overhead, bound = 3×4+1+2 = 15 seconds. Three timeouts exhaust the budget; no fourth attempt is started. Only declared timeout/transient categories retry in this local model; authentication, permission and invalid-request failures stop.

  Separate read-only picture: two attempts with 1-second caps and zero gap; first timeout 1 second plus supplied success .4 second totals 1.4 seconds. Cancellation before attempt 2/backoff stops future work at 1 second. Helper validates finite nonnegative numeric durations excluding bool, exact integer attempt count and one fewer gap than attempts.

  Code adds virtual durations only, with cancellation at supplied between-attempt boundaries. Caller timeout/cancellation does not prove remote work stopped or undo committed effects. No real sleep, network, clock measurement or asynchronous cancellation is implemented.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main virtual policy: at most 3 total attempts, each capped at 4 seconds of caller waiting, with gaps 1 and 2 seconds before attempts 2 and 3. Under enforced local caps and excluding overhead, bound = 3×4+1+2 = 15 seconds. Three timeouts exhaust the budget; no fourth attempt is started. Only declared timeout/transient categories retry in this local model; authentication, permission and invalid-request failures stop.

  Separate read-only picture: two attempts with 1-second caps and zero gap; first timeout 1 second plus supplied success .4 second totals 1.4 seconds. Cancellation before attempt 2/backoff stops future work at 1 second. Helper validates finite nonnegative numeric durations excluding bool, exact integer attempt count and one fewer gap than attempts.

  Code adds virtual durations only, with cancellation at supplied between-attempt boundaries. Caller timeout/cancellation does not prove remote work stopped or undo committed effects. No real sleep, network, clock measurement or asynchronous cancellation is implemented.

- Three attempts each time out after 4 seconds with gaps 1 and 2. A fourth is proposed. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main virtual policy: at most 3 total attempts, each capped at 4 seconds of caller waiting, with gaps 1 and 2 seconds before attempts 2 and 3. Under enforced local caps and excluding overhead, bound = 3×4+1+2 = 15 seconds. Three timeouts exhaust the budget; no fourth attempt is started. Only declared timeout/transient categories retry in this local model; authentication, permission and invalid-request failures stop.

  Separate read-only picture: two attempts with 1-second caps and zero gap; first timeout 1 second plus supplied success .4 second totals 1.4 seconds. Cancellation before attempt 2/backoff stops future work at 1 second. Helper validates finite nonnegative numeric durations excluding bool, exact integer attempt count and one fewer gap than attempts.

  Code adds virtual durations only, with cancellation at supplied between-attempt boundaries. Caller timeout/cancellation does not prove remote work stopped or undo committed effects. No real sleep, network, clock measurement or asynchronous cancellation is implemented.

- A report omits this qualification: A **timeout** means the caller lacks a timely result, not necessarily that the remote operation did not occur.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main virtual policy: at most 3 total attempts, each capped at 4 seconds of caller waiting, with gaps 1 and 2 seconds before attempts 2 and 3. Under enforced local caps and excluding overhead, bound = 3×4+1+2 = 15 seconds. Three timeouts exhaust the budget; no fourth attempt is started. Only declared timeout/transient categories retry in this local model; authentication, permission and invalid-request failures stop.

  Separate read-only picture: two attempts with 1-second caps and zero gap; first timeout 1 second plus supplied success .4 second totals 1.4 seconds. Cancellation before attempt 2/backoff stops future work at 1 second. Helper validates finite nonnegative numeric durations excluding bool, exact integer attempt count and one fewer gap than attempts.

  Code adds virtual durations only, with cancellation at supplied between-attempt boundaries. Caller timeout/cancellation does not prove remote work stopped or undo committed effects. No real sleep, network, clock measurement or asynchronous cancellation is implemented.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Two attempts, each up to 5 seconds, gap 2. Compute bound and decide whether authentication failure should retry.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled **timeout**s retries and **cancellation evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **timeout**s retries and **cancellation**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Timeout**s retries and **cancellation**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Timeout**s retries and **cancellation**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Two attempts, each up to 5 seconds, gap 2. Compute bound and decide whether authentication failure should retry.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

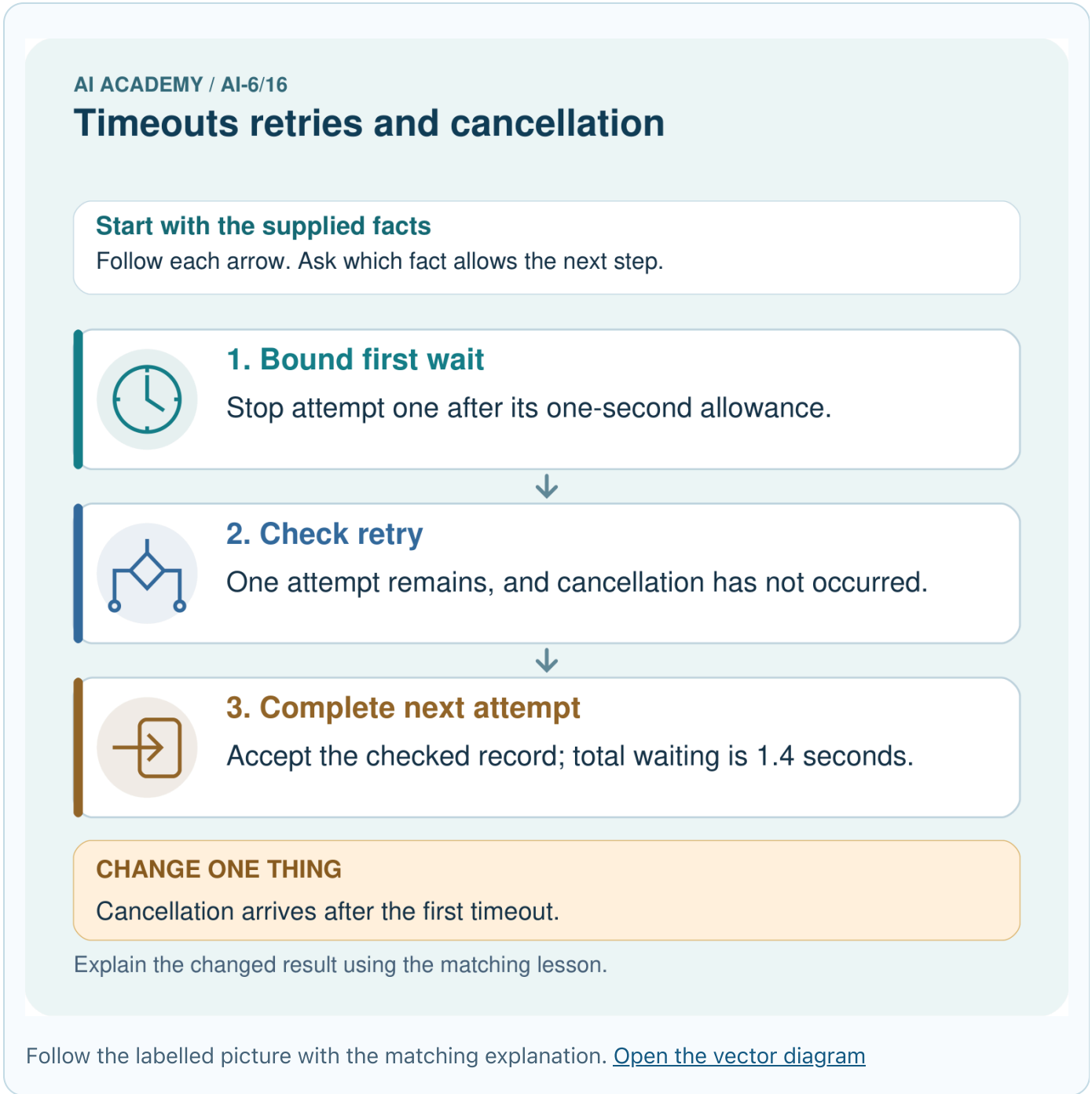
- Retry within declared limits. A read-only lookup allows two attempts of at most one second each. **Cancellation** forbids further attempts.

- First attempt times out at 1 second. Second succeeds in 0.4 seconds. Ignore overhead. No **cancellation** occurs.

- Calculate waiting time for a one-second **timeout** followed by a 0.4-second success. What changes if **cancellation** arrives after the first attempt but before the retry?

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Retry within declared limits. A read-only lookup allows two attempts of at most one second each. **Cancellation** forbids further attempts.
- Which labelled step matches this detail? Accept the checked record; total waiting is 1.4 seconds.
- 1. Complete next attempt
- 2. Bound first wait

- 3. Check retry

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Complete next attempt Accept the checked record; total waiting is 1.4 seconds. A **timeout** bounds waiting; a retry tries again; **cancellation** withdraws the request.

They are separate conditions on the next action.

Changed-case question and optional hint

- Changed-case practice: **Cancellation** arrives after the first **timeout**. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. Do not start the second attempt.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Do not start the second attempt. **Cancellation** overrides the unused retry allowance.

Repair and retain the record

Review Approval gates and idempotency (ai-6/15). Goal: Explain the **mechanism** and terminology of approval gates and idempotency. Short practice: Trace the two deliveries of approved K9.

How many reservations should occur, and why can a new K10 for desk 5 not borrow K9's approval?. Return to this week and explain what you changed.

Review State machines for bounded agents (ai-6/13). Goal: Explain the **mechanism** and terminology of state machines for bounded agents. Short practice: Trace A and B through the state diagram.

Name B's forbidden transition and explain whether adding checking→querying permits that shortcut.. Return to this week and explain what you changed.

Review Evaluation pipelines (ai-6/10). Goal: Explain the **mechanism** and terminology of evaluation pipelines. Short practice: Calculate A's and B's correct rates on all four cases.

Identify their shared failure and explain why removing that case after scoring changes the honesty of the comparison.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

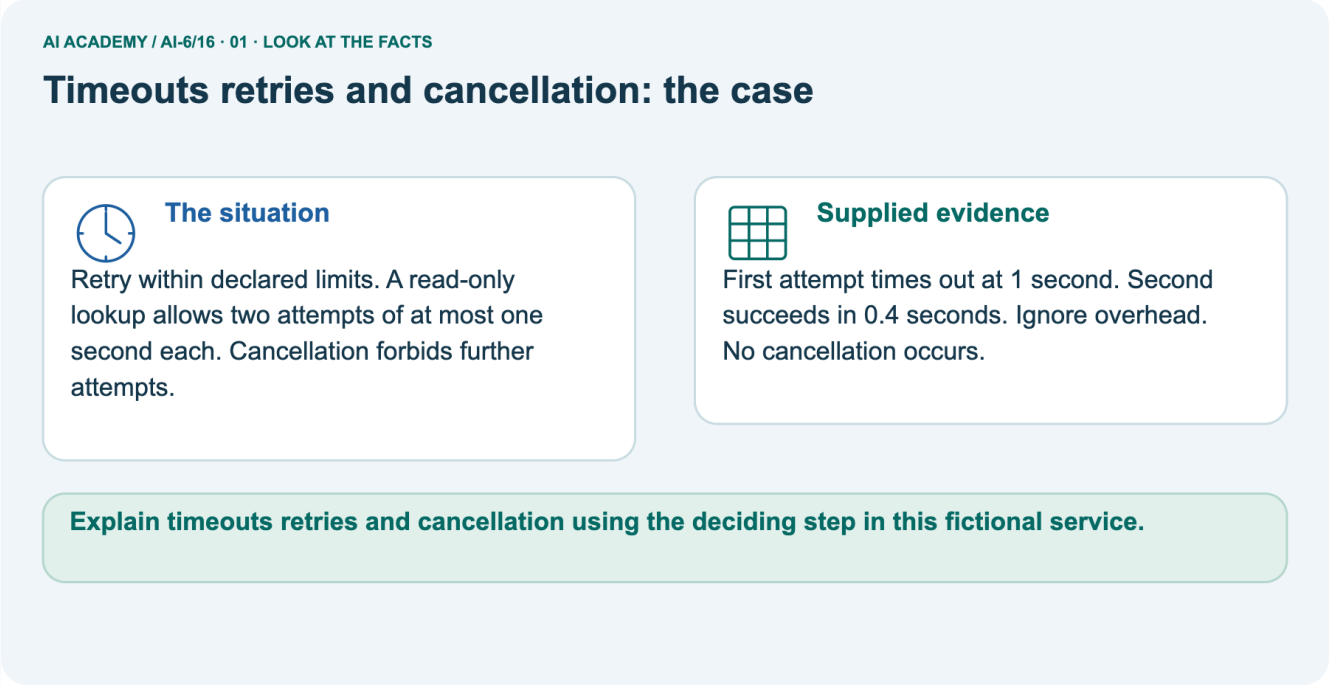
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Timeouts retries and cancellation: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

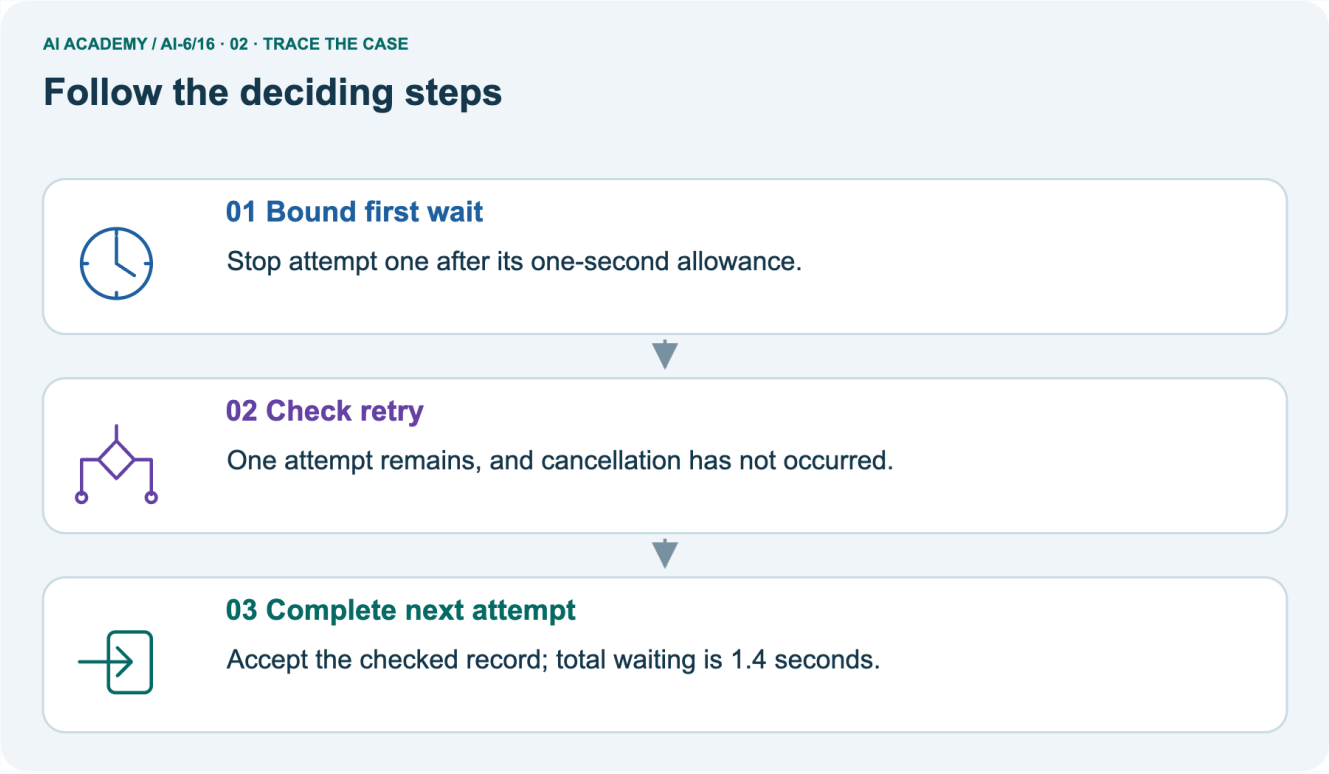
Retry within declared limits. A read-only lookup allows two attempts of at most one second each. **Cancellation** forbids further attempts. Supplied facts: First attempt times out at 1 second. Second succeeds in 0.4 seconds.

Ignore overhead. No **cancellation** occurs.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Bound first wait: Stop attempt one after its one-second allowance. → Check retry: One attempt remains, and **cancellation** has not occurred. → Complete next attempt: Accept the checked record; total waiting is 1.4 seconds.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Timeouts retries and cancellation — evidence trace

AI ACADEMY / AI-6/16 · 03 · INSPECT THE LESSON MODEL

Timeouts retries and cancellation — evidence trace

Input and scope	Check or calculation	Conclusion and limit
Three attempts each time out after 4 seconds with gaps 1 and 2. A fourth is proposed.	Retry only declared transient failures. Use at most three attempts with waiting gaps 1 and 2 seconds; the caller waits at most 4 seconds for each attempt under the enforced local timeout model. Worst-case elapsed bound excluding overhead is 3*4+1+2=15 seconds.	Budget is exhausted after 15 seconds under the model; reject fourth attempt. A late side-effect result must still be reconciled.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

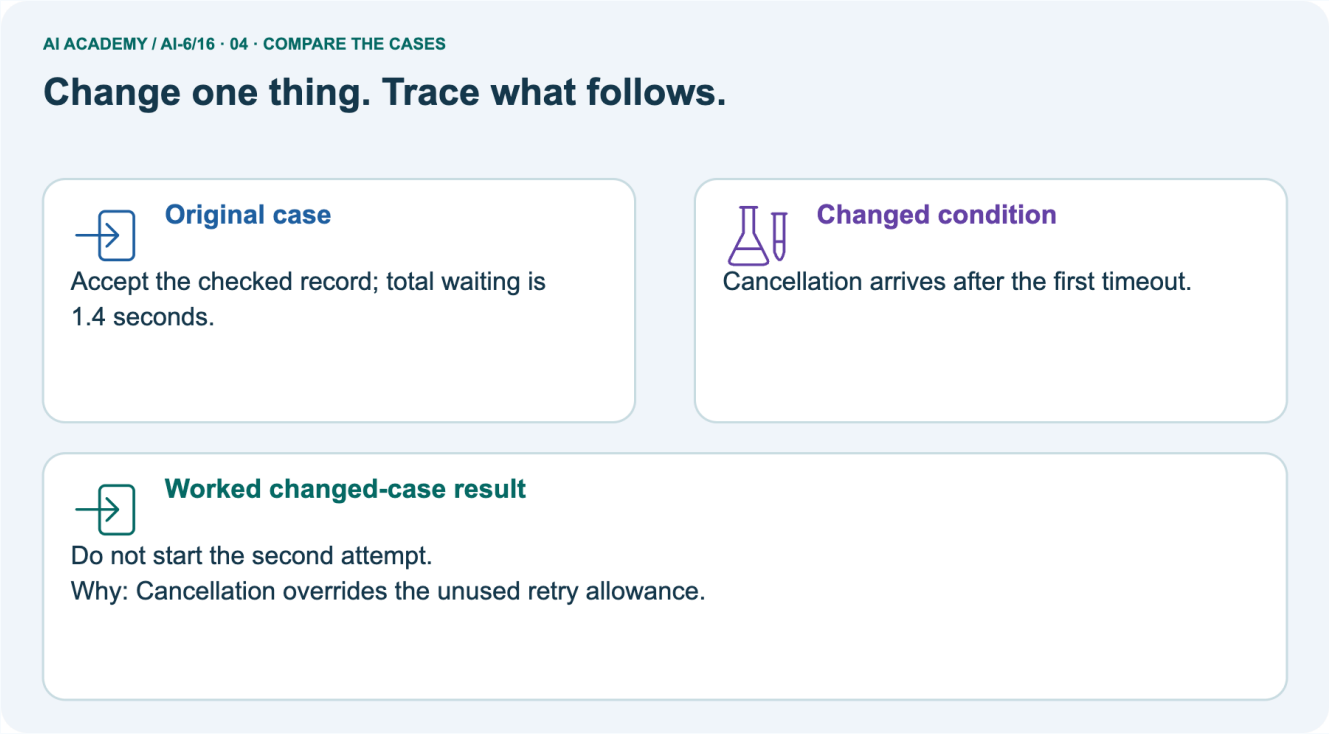
Follow the labelled picture with the matching explanation. Open the vector diagram

Timeouts retries and **cancellation** — **evidence** trace. Three attempts each time out after 4 seconds with gaps 1 and 2. A fourth is proposed. | Retry only declared transient failures. Use at most three attempts with waiting gaps 1 and 2 seconds; the caller waits at most 4 seconds for each attempt under the enforced local **timeout** model. Worst-case elapsed bound excluding overhead is 3*4+1+2=15 seconds. | Budget is exhausted after 15 seconds under the model; reject fourth attempt. A late side-effect result must still be reconciled.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

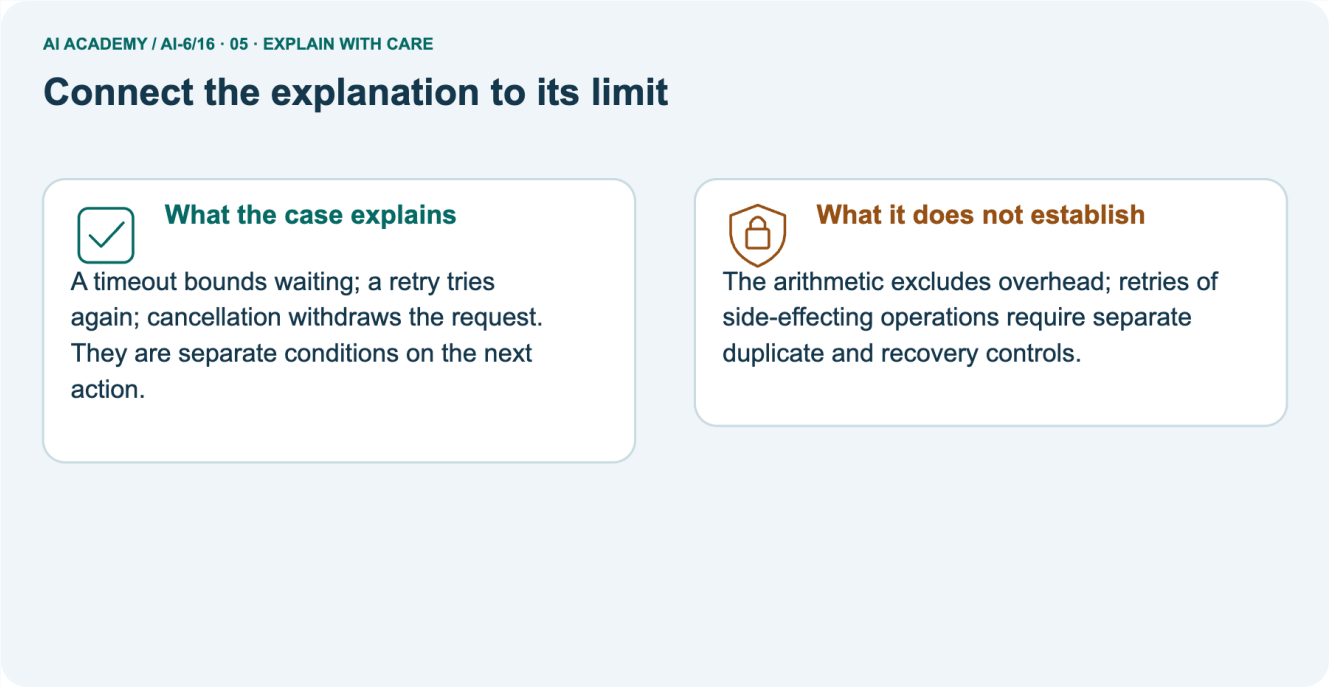
Original case: Accept the checked record; total waiting is 1.4 seconds. Changed condition: **Cancellation** arrives after the first **timeout**. Expected result: Do not start the second attempt.

Reason: **Cancellation** overrides the unused retry allowance.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

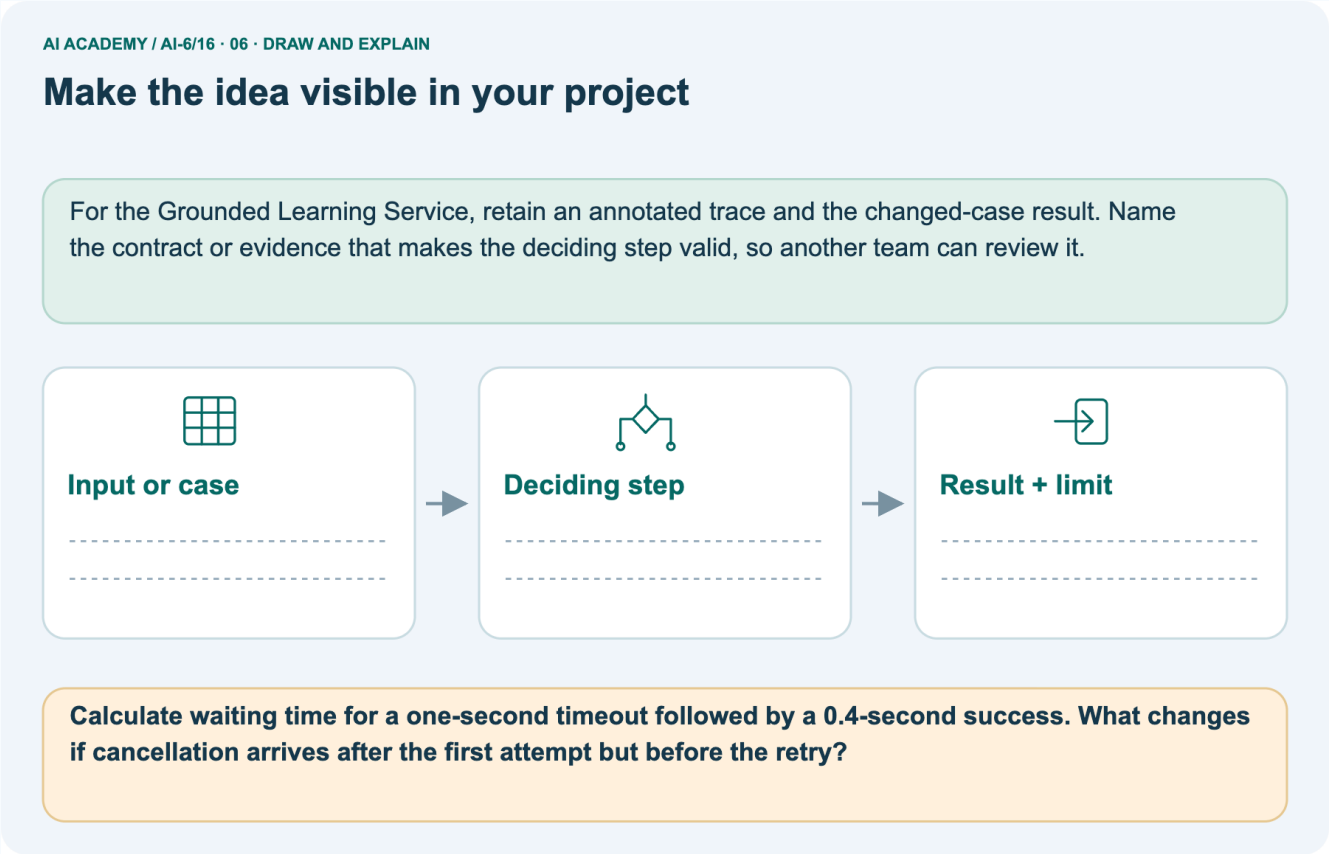
A **timeout** bounds waiting; a retry tries again; **cancellation** withdraws the request. They are separate conditions on the next action.

Limit: The arithmetic excludes overhead; retries of side-effecting operations require separate duplicate and recovery controls.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate waiting time for a one-second **timeout** followed by a 0.4-second success.

What changes if **cancellation** arrives after the first attempt but before the retry?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Timeouts retries and cancellation: the case

02. Follow the deciding steps

03. Timeouts retries and cancellation — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.